phensim: genotype, phenotype and summary-statistic simulation for statistical genetics

Technical report

phensim 1.0.0: API and numerical contracts

October 2026

Abstract

phensim is a NumPy-only Python package for simulating the data that statistical-genetics methods are evaluated on. It covers genotypes with realistic linkage disequilibrium (LD), phenotypes with known genetic architecture, GWAS summary statistics under population LD, finite reference panels, pedigrees and their relationship matrices, kinship estimators and PLINK output. Every simulator is seeded, validates its inputs, and returns its ground truth alongside the data.

The package contains a built-in, JIT-compiled implementation of Hudson's coalescent with recombination. Across 30 seeds it agrees with msprime on site counts, nucleotide diversity and the decay of LD with distance. The trait simulators draw the infinitesimal background from the same scaled genomic relationship matrix a mixed model fits, through an exact factor that never forms an $n \times n$ matrix. The summary-statistic layer reproduces, bit for bit, the benchmark simulators of the LDpred3 family that phensim now hosts.

We describe the models, the algorithms and the numerical contracts, and report measured agreement between realized and target quantities together with indicative costs. The API descriptions include subsequent additions and repairs; all measured tables and figures retain the historical evidence snapshot from phensim 1.0.0.dev1 at revision `a7371c5`. Those measurements have not been rerun for the current implementation.

1 Introduction

Simulation is how a statistical-genetics method earns trust before it meets real data: only with a known truth can bias, calibration, power and cost be measured. Methods that model LD (polygenic scores, fine-mapping, heritability and genetic-correlation estimators) need simulators that get LD, allele frequencies and the generating model right. Their summary-statistic variants additionally need the regression-with-summary-statistics (RSS) model [14] and realistic reference-panel noise.

phensim began as a teaching snippet. It grew into the shared simulator of a family of packages (ldpred3, bipred, gwfm, g2gfun, multipgs, and the pedigree code of ltpred) whose benchmark suites previously each carried their own copies. It follows six design principles:

- **Model consistency.** Data are generated from the model an estimator fits: the infinitesimal background uses the same scaled GRM a mixed model uses, and summary statistics use the exact RSS likelihood.
- **Truth returned.** Every simulator returns its generating components (background, QTL effects, causal sets, liabilities, the relationship matrix), so estimands can be scored exactly.
- **Validated inputs.** Malformed inputs raise before any work is done or any file is written, rather than being coerced.
- **Reproducibility.** Every function takes a seed or a NumPy `Generator`, and draw orders are documented contracts.

- **Minimal dependencies.** The core needs only NumPy. Numba accelerates the coalescent; msprime is an optional second backend.
- **Memory discipline.** Genotypes are `int8`, LD is validated in bounded tiles, and the large draws avoid $n \times n$ or genome-wide intermediates.

2 Package overview

Table 1 lists the public API. Discrete genotype simulators return sample-major `int8` dosages in $\{0, 1, 2\}$ with columns in physical order, or phased alleles when requested. Gaussian predictors are continuous `float64` draws. LD is passed as a list of dense (R, ix) blocks that tile the variants exactly once. Figure 1 contrasts the LD of the three structured genotype models.

Table 1: Public API by module.

Module	Functions
genotypes	`simulate_independent`, `simulate_population_structure`, `simulate_ar1_blocks`, `realistic_block_sizes`, `simulate_haplotype_blocks`, `simulate_coalescent`, `simulate_by_mutation_rate`
phenotypes	`simulate_trait`, `simulate_binary_trait`, `simulate_confounded_trait`, `simulate_gxe_trait`, `simulate_correlated_traits`, `ascertain_case_control`, `h2_liability`, `n_eff_case_control`
ancestry	`drift_frequencies`, `simulate_populations`, `simulate_admixed`, `simulate_split_coalescent`, `simulate_genome`, `simulate_meta_sumstats`
hapnest	`simulate_hapnest`, `iter_hapnest`
sumstats	`simulate_effects`, `simulate_effects_pair`, `genetic_correlation`, `simulate_sumstats`, `simulate_sumstats_pair`, `shake_ld`, `prepare_blocks`, `simulate_gaussian_individuals`, `gwas_scan`
pedigree	`simulate_pedigree`, `pedigree_birth_times`, `kinship_from_pedigree`, `mendelian_draw`
kinship	`grm`, `ibs_kinship`, `loco_kinships`, `iter_loco_kinships`, `windowed_kinships`
io	`write_plink`

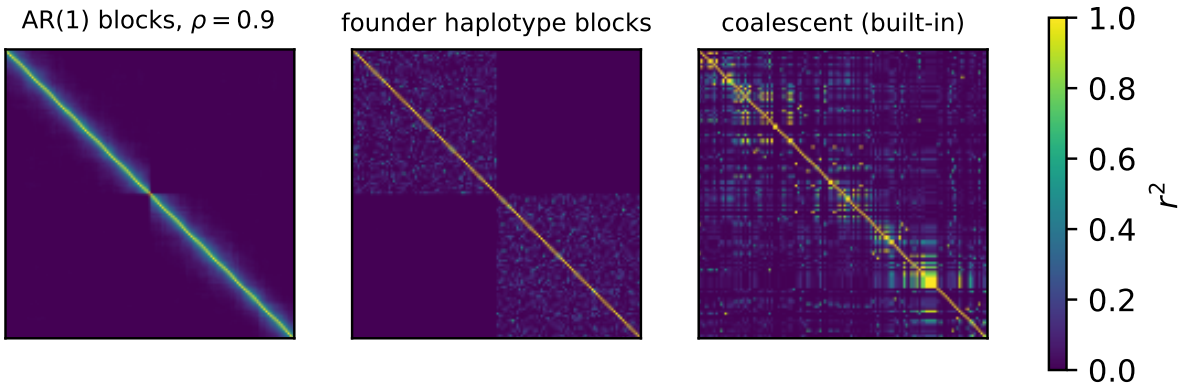


Figure 1: Squared correlation between 120 consecutive SNPs in 1000 simulated individuals. AR(1) blocks have smooth geometric decay and hard block edges. Founder haplotype copying produces patchy haplotype sharing. The coalescent produces recombination-driven, irregular LD whose block boundaries are not imposed.

3 Genotype models

Independent and structured SNPs. `simulate_independent` draws $G_{ij} \sim \mathrm{Bin}(2, p_j)$, where p_j is either a fixed minor allele frequency or drawn from a named frequency spectrum (Beta, uniform, rare or common). Each site's minor allele is the counted allele with probability 1/2.

`simulate_population_structure` draws base frequencies

$$p_j = \mathrm{clip}\big(\mathrm{maf} + N(0, 0.05^2),\, 0.05,\, 0.95\big)$$

and then, independently for each of K populations, either the normal approximation $p_{jk} \sim N(p_j, F p_j(1-p_j))$ or the exact Balding–Nichols distribution [1]

$$p_{jk} \sim \mathrm{Beta}\Big(\tfrac{p_j(1-F)}{F},\, \tfrac{(1-p_j)(1-F)}{F}\Big),$$

clipped to $[0.01, 0.99]$.

AR(1) haplotype blocks. Within a block of k SNPs, each individual carries two latent haplotypes $z \sim N(0, C)$ with $C_{ij} = \rho^{|i-j|}$. A haplotype carries the allele at SNP j when $z_j > \Phi^{-1}(1-f_j)$, so the allele frequency is exactly f_j; the dosage sums the two haplotypes. The default draws $z = L\varepsilon$ with $LL^\top = C + 10^{-8} I$. This is the family's historical stream, which phensim reproduces bit for bit. The alternative recursion $z_j = \rho z_{j-1} + \sqrt{1-\rho^2}\,\varepsilon_j$ costs $O(nk)$, needs no factorization, and is exact at $\rho = \pm 1$. Both consume the same normals. `realistic_block_sizes` draws log-normal block lengths with a chosen coefficient of variation, which stresses the few large blocks that dominate quadratic LD work.

Founder haplotype copying. Each block has a small set of Bernoulli(0.3) founder haplotypes. Every individual copies two founders, with per-site flips at a mutation rate. This gives genuine haplotype sharing at the cost of one pass.

Coalescent genotypes. `simulate_coalescent` grows a segment until at least m SNPs pass the MAF filter. It starts at the larger of 1 Mb and $m/1200$ Mb and multiplies the length by 1.8 per retry. It then keeps the first m of those SNPs in physical order and cuts them into contiguous blocks. `simulate_by_mutation_rate` instead fixes the segment and recombination rate, so the genealogy is determined by the seed and the mutation rate alone controls SNP density. This is the density lever used in the family's scaling benchmarks.

4 The built-in coalescent engine

The engine is a compact implementation of Hudson's coalescent with recombination [7] for one constant-size diploid population, written to compile end to end under Numba. It writes dosages directly instead of building a tree-sequence table collection.

Ancestry. The simulation starts from $2n$ haploid lineages carrying $[0, L)$. With k lineages, the next event occurs after an exponential waiting time at total rate

$$\lambda = \binom{k}{2}\frac{1}{2N_e} + r\sum_i \ell_i,$$

where ℓ_i is the distance between lineage i's leftmost and rightmost ancestral positions. A recombining lineage is chosen in proportion to ℓ_i through a Fenwick tree [4], in $O(\log k)$. The breakpoint is uniform on that span, gaps included, and is clamped to the representable interior of the span.

Coalescence merges the two segment lists. Each overlapping interval creates a node and two edges, and each segment carries the number of samples below it. An interval whose count reaches $2n$ has found its most recent common ancestor and leaves the simulation, which is what makes the process terminate.

Mutations and genotypes. Each edge receives Poisson($\mu\, t_{\text{branch}}\,\text{span}$) infinite-sites mutations at uniform positions. Densification then sweeps the marginal trees left to right, inserting edges by (left, parent time) and removing them by (right, −parent time), as in tskit. Each mutation adds one to every sample below its node, and haplotypes $2i$ and $2i+1$ form individual i.

Engineering. All work arrays are preallocated from expected event counts. On overflow the engine doubles the buffer and reruns with the same seed, which reproduces the same events, so a caller never sees a partial result. The ancestry seed is the caller's seed masked to 31 bits, and the mutation stream uses $(s \cdot 2654435761) \bmod 2^{31}$. The public functions accept seeds in $[1, 2^{31})$, the range both backends honour. Without Numba the same code runs as pure Python on NumPy's global generator, whose state is saved and restored around each call. Linked-list and Fenwick-weight invariants are asserted at every event boundary in the test suite.

Agreement with msprime. msprime [9, 2] is the reference implementation; phensim calls it on a continuous genome with the binary mutation model. We simulated $n = 100$ diploids on 200 kb with $N_e = 10^4$ and $r = \mu = 10^{-8}$, for 30 seeds per backend.

Table 2 shows that site counts and nucleotide diversity agree with each other and with the neutral expectations $\mathbb{E}[S] = \theta \sum_{i<2n} 1/i$ and $\mathbb{E}[\pi] = \theta$, where $\theta = 4N_e\mu L$. Figure 2 shows the decay of r^2 with physical distance: in no distance bin do the backends differ by more than 1.6 standard errors. These are bounded distributional checks, not a proof of equivalence; the backends draw different events from the same model.

Table 2: Coalescent backends, mean over 30 seeds (standard error).

Backend	Segregating sites (SE)	Diversity $\sum 2f(1-f)$ (SE)
built-in (Numba)	478 (9)	82 (2)
msprime 1.4	477 (6)	82 (2)
neutral expectation	470	80

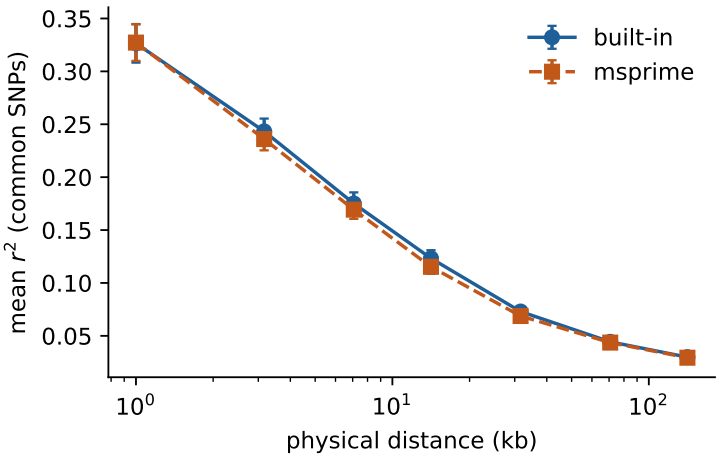


Figure 2: Mean r^2 between common SNPs (MAF ≥ 0.05) by physical distance, for the built-in engine and msprime with the same design as Table 2. Bars are 95% intervals over seeds.

5 Kinship

`grm` standardizes each variant over its called samples (missing calls contribute zero) [12] and forms $ZZ^\top/m$. It then applies the EMMAX scaling [8]: subtract the mean off-diagonal element, then

divide by the mean diagonal.

Leave-one-chromosome-out matrices subtract each chromosome's Gram from the full Gram under the same global standardization, so they are exact rather than re-estimated. `windowed_kinships` splits the Gram the same way into a local window and its complement. `ibs_kinship` is the per-pair fraction of identical called genotypes, computed with one-hot GEMMs.

6 Phenotype models

`simulate_trait` builds $\ell = u + q + e$. The infinitesimal background is $u \sim N(0, \sigma_u^2 K)$, where K is the scaled GRM a mixed model would fit. The QTL component q comes from causal variants, and the residual is $e \sim N(0, 1 - h^2)$. Three architectures split h^2: `mixed` gives half to u and half to q, `infinitesimal` gives all of it to u, and `qtl` gives all of it to q. QTL effects are normal or of equal magnitude, act on standardized causal columns, and are rescaled so that the in-sample $\mathrm{Var}(q)$ equals its target exactly.

Proposition 1 (Matrix-free background)**.** *Let Z be the called-standardized genotypes, whose columns sum to zero, and let $S = \sum_{ij} Z_{ij}^2$. The EMMAX-scaled GRM equals $K = \frac{n-1}{S} ZZ^\top + \frac{1}{n} J$. Hence*

$$u = \sigma_u \Big(\sqrt{\tfrac{n-1}{S}}\, Zw + \tfrac{c}{\sqrt{n}} \Big), \qquad w \sim N(0, I_m),\ c \sim N(0,1),$$

has covariance exactly $\sigma_u^2 K$.

Proof. Write $K_0 = ZZ^\top/m$. Its grand sum is $\mathbf{1}^\top ZZ^\top \mathbf{1}/m = 0$ and its trace is S/m. The mean off-diagonal element is therefore $-S/(mn(n-1))$. After subtracting it, the mean diagonal is $S/(m(n-1))$, and dividing by it gives the stated K. The factor $F = [\sqrt{(n-1)/S}\, Z,\ \mathbf{1}/\sqrt{n}]$ satisfies $FF^\top = K$. □

The draw therefore needs $m + 1$ normals and one matrix–vector product, with no $n \times n$ matrix and no eigendecomposition. Its in-sample variance has expectation $\sigma_u^2(1 - 1/n)$, because $\mathbf{1}^\top K \mathbf{1} = n$. A supplied K is drawn through its eigendecomposition instead, and the two give the same distribution: in Table 3 the background variances are 0.504 and 0.512 against an expectation of 0.500.

Derived traits.

- **Binary traits** threshold the liability at $\Phi^{-1}(1 - K)$.
- **Confounded traits** add a share s of variance along the leading eigenvector of K, the canonical population-structure axis, reusing the same eigendecomposition for the background.
- **Gene–environment traits** add standardized causal genotypes multiplied by a standardized environment, scaled to the interaction variance.
- **Correlated traits** share both components across traits: $g_b \propto r_g u_1 + \sqrt{1 - r_g^2}\, u_2 + r_g q_1 + \sqrt{1 - r_g^2}\, q_2$. This targets the *total* genetic correlation and is exact at $r_g = \pm 1$.

`ascertain_case_control` samples exact case and control counts, and `h2_liability` implements the observed-to-liability transformation of Lee et al. [10]. It warns unless the study case fraction is given.

Table 3 compares realized and target quantities over 40 draws on a coalescent panel of 1500 individuals and 3000 SNPs. Every target is recovered within simulation error. The realized variance shares are fractions of the realized liability variance, so they fluctuate with the sample covariances between components.

Table 3: Phenotype simulators: target versus realized. For `simulate_trait` the three numbers are the background, QTL and total genetic shares of liability variance.

Simulator and quantity	Target	Realized mean (SD), 40 draws
`simulate_trait` (mixed)	0.25 / 0.25 / 0.50	0.25 (0.03) / 0.25 (0.02) / 0.50 (0.03)
`simulate_trait` (infinitesimal)	0.50 / 0.00 / 0.50	0.50 (0.04) / 0.00 (0.00) / 0.50 (0.04)
`simulate_trait` (qtl)	0.00 / 0.50 / 0.50	0.00 (0.00) / 0.51 (0.01) / 0.51 (0.01)
background variance, K omitted / supplied	0.500	0.504 (0.074) / 0.512 (0.081)
`simulate_binary_trait` prevalence	0.05	0.049 (0.007)
`simulate_binary_trait` prevalence	0.20	0.201 (0.012)
`simulate_correlated_traits` r_g	0.00	-0.023 (0.136)
`simulate_correlated_traits` r_g	0.50	0.476 (0.095)
`simulate_correlated_traits` r_g	0.90	0.896 (0.016)
`simulate_gxe_trait` interaction share	0.20	0.199 (0.012)

7 Summary statistics

The RSS model. With standardized effects β and block-diagonal population LD R, marginal GWAS effects are drawn as [14]

$$\hat{\beta} = R\beta + \varepsilon, \qquad \varepsilon \sim N(0, R/n),$$

or $N(0, DRD)$ with $D_{jj} = n_j^{-1/2}$ for per-variant sample sizes. Each block contributes $\varepsilon = Fz/\sqrt{n}$ with $z \sim N(0, I_r)$. The factor F is chosen per call:

- by default, the exact factor of R: Cholesky, or $V\Lambda^{1/2}$ for singular LD, with no diagonal noise added;
- with `jitter`, the Cholesky factor of $R + \epsilon I$;
- with `factors`, matrices supplied by the caller, for example an eigenvalue-clipped root of thresholded LD, which need not be PSD.

The signal always uses R. `prepare_blocks` validates and factors the blocks once into a read-only snapshot that every consumer accepts.

Effects. `simulate_effects` draws a sparse, polygenic, equal-magnitude or MAF-dependent shape and rescales it so that $\beta^\top R\beta = h^2$ exactly. The MAF-dependent shape uses $\beta_j \propto [2f_j(1-f_j)]^{(1+\alpha)/2}$, so the per-allele variance is proportional to $[2f_j(1-f_j)]^\alpha$. This is the convention of LDAK [11], SBayesS [13] and ldpred3, in which $\alpha = -1$ means no MAF dependence.

`simulate_effects_pair` draws shared effects with correlation ρ and scales each trait to its own h^2. The causal layout is either a shared Bernoulli(p) set, or exact per-trait and shared counts (n_a, n_b, n_{ab}); the latter is the bivariate causal mixture of MiXeR [5], whose target genetic correlation is $\rho\, n_{ab}/\sqrt{n_a n_b}$. `genetic_correlation` returns the correlation realized under the LD.

Two studies and reference panels. `simulate_sumstats_pair` draws z_a and then z_b per block and sets $\varepsilon_b = F(\rho z_a + \sqrt{1-\rho^2} z_b)/\sqrt{n_b}$, which gives cross-covariance $\rho R/\sqrt{n_a n_b}$. Here ρ is the sampling-noise correlation, not the fraction of overlapping participants.

`shake_ld` simulates a reference panel: it draws $X = ZF^\top$ with n_{ref} rows, standardizes the columns, and returns $(1-s)X^\top X/n_{\text{ref}} + sI$. An optional chunked path accumulates the same draws with the stable one-pass update of Chan et al. [3], bounding memory at $O(\text{chunk} \cdot k + k^2)$.

Individual-level GWAS. `gwas_scan` fits each variant on its called samples. It reports the Pearson correlation r, the standard error $\sqrt{(1-r^2)/(n-2)}$, the OLS t statistic and a normal p-value. A variant whose called genotypes or called phenotypes are constant, tested exactly, returns NaN instead of a roundoff statistic.

Validation tolerances. Blocks must tile the variants and be finite, symmetric, unit-diagonal, bounded in $[-1, 1]$ and PSD. Validation runs in tiles of 256 rows, so genome-wide LD is never assembled. Tolerances follow the input precision: the relative tolerance is $\max(10^{-7}, 4\epsilon)$, and negative eigenvalues are clipped up to $k \max(64\,\epsilon_{64} \max(1, \lambda_{\max}), \epsilon_{\text{in}})$. float32 LD, the family's usual storage, is therefore judged at float32 roundoff.

Table 4 confirms each property against its theoretical expectation. The empirical noise covariance matches R/n to the error expected of a Wishart estimate. Panel noise matches the large-sample standard deviation of a sample correlation, $(1-\rho^2)/\sqrt{n_{\text{ref}}}$. Null scans are calibrated: mean $\lambda_{\text{GC}} = 1.002$ (SD 0.061) over 40 null traits with 5% missing calls.

Table 4: Summary-statistic layer: properties versus expectation.

Check	Expected	Realized
`simulate_effects`: $\lvert\beta^\top R\beta - h^2\rvert$, 4 architectures	0	1.1e-16
`simulate_sumstats`: $\lVert n\,\widehat{\text{Cov}} - R\rVert_F / \lVert R\rVert_F$, 4000 draws	≈ 0.096	0.095
`simulate_sumstats_pair`: noise correlation	0.40	0.399
`simulate_effects_pair`: mean realized r_g (40/40 causal, 20 shared, $\rho = 0.8$)	0.40	0.395 (0.160)
`shake_ld`: off-diagonal RMSE, $n_{\text{ref}} = 100$	≈ 0.096	0.097
`shake_ld`: off-diagonal RMSE, $n_{\text{ref}} = 500$	≈ 0.043	0.045
`shake_ld`: off-diagonal RMSE, $n_{\text{ref}} = 2500$	≈ 0.019	0.020
`gwas_scan`: λ_{GC}, 40 null traits, 5% missing calls	1	1.002 (0.061)

Parity with the family. With `jitter=1e-4`, per-trait sample sizes and `shrink`, these functions reproduce byte for byte:

- ldpred3: `_metrics.sumstats` and `_realistic_ld.panel_genome`;
- bipred: `rg_architectures.sim_effects`, `sumstats_pair`, `ref_panel` and `realized_rg`, and `mixer_overlap._sim_mixture`.

This was checked against the siblings' own source and is pinned by verbatim oracles in `tests/test_family_parity.py`. A sibling can therefore replace its local copy without changing any seeded result.

8 Pedigrees

`simulate_pedigree` produces multi-generation trio tables from founder couples. Each couple has two to four children, and remarriage produces half-siblings. `kinship_from_pedigree` orders individuals parents first and fills the additive relationship matrix with Henderson's tabular recursion [6]:

$$A_{kj} = \tfrac{1}{2}(A_{s_k j} + A_{d_k j}), \qquad A_{kk} = 1 + \tfrac{1}{2} A_{s_k d_k},$$

where unknown parents contribute zero. The cost is $O(n^2)$, and the result is exactly symmetric and includes inbreeding.

`mendelian_draw` samples $a \sim N(0, A)$ in $O(n)$ storage: $a_i = \sum_p a_p/2 + \sqrt{w_i} z_i$, summing over known parents, with $w_i = 1 - \sum_p A_{pp}/4$. This is the Mendelian sampling variance $(1 - (F_s + F_d)/2)/2$ when both parents are known. The required diagonals come from a memoized pair recursion.

Over 2000 draws on a simulated 367-person pedigree, which includes inbred individuals, the empirical covariance matched A with a standardized root-mean-square error of 0.998; pure sampling noise gives 1. `pedigree_birth_times` assigns generation-coherent birth years and rejects inconsistent pedigrees.

9 Output

`write_plink` writes SNP-major PLINK 1 binary files. Genotypes are coded 00, 10 and 11 for zero, one and two copies of BIM allele 2, with 01 for missing, packed four samples per byte. Every genotype and metadata check, including unique and printable sample identifiers, PLINK chromosome codes and integral positions, completes before any file is opened.

10 Performance

Table 5 gives indicative single-thread costs. These were measured on a shared machine (Darwin arm64, load average 11.5), so absolute numbers will vary; the comparisons are what matter.

- At this size the msprime backend is somewhat faster than the built-in engine. The built-in engine's value is that it has no compiled dependency and writes genotypes directly.
- The matrix-free trait draw is about 25 times faster than drawing through a supplied K. The gap widens with n, because the eigendecomposition costs $O(n^3)$ while the matrix-free draw costs $O(nm)$.
- Prepared LD blocks remove the per-call validation and factorization, which dominate repeated summary-statistic draws.
- The AR(1) recursion is not faster at a block size of 200; it matters for very large blocks and exact $\rho = \pm 1$.

Table 5: Indicative wall-clock cost, median of three runs after a warm-up, BLAS pinned to one thread.

Call	Seconds (median of 3)
simulate_coalescent, $n = 1000$, $m = 10^4$, built-in	0.73
simulate_coalescent, $n = 1000$, $m = 10^4$, msprime	0.47
simulate_ar1_blocks, $n = 5000$, $m = 2 \times 10^4$, Cholesky	2.51
simulate_ar1_blocks, $n = 5000$, $m = 2 \times 10^4$, scan	3.51
simulate_trait, $n = 3000$, $m = 5000$, K omitted (matrix-free)	0.22
simulate_trait, $n = 3000$, $m = 5000$, K supplied (eigh)	5.37
grm, $n = 3000$, $m = 5000$	0.43
simulate_sumstats, $m = 5 \times 10^4$ (250 blocks of 200), raw blocks	0.15
simulate_sumstats, same, prepared blocks	0.01
shake_ld, same, $n_{\text{ref}} = 500$	0.29

11 Reproducibility and integration

Seeds. Seeds are integers, `None`, or NumPy `Generator`s. A `Generator` continues its stream, so composed simulations remain reproducible.

Later simulation APIs. `drift_frequencies` exposes per-population Balding–Nichols or normal drift. `simulate_populations` draws population-specific AR(1) blocks; `simulate_admixed` joins ancestry tracts along a genetic map and returns per-haplotype ancestry. `simulate_split_coalescent` adds a simultaneous population split and an optional admixture pulse through msprime. `simulate_genome` seeds independent coalescent chromosomes, and `simulate_meta_sumstats` combines independent population GWAS by inverse-variance weighting. These are described in `docs/technical.md`.

`simulate_hapnest` copies phased references using Gamma segment ages and exponential lengths in cM, retaining only alleles older than the segment. It supports discrete donor populations, optional Numba, sample batches and disk-backed output; `iter_hapnest` yields batches directly. Its model and conventions are detailed in `docs/hapnest.md`. `simulate_gaussian_`

`individuals` draws continuous predictors with population covariance R, including singular support, without sample rescaling.

Numerical boundaries. Fenwick sampling is bounded to active lineage slots; mutation positions remain inside half-open edge intervals after rounding. Constant called dosage columns have exactly zero GRM weight. Birth times solve relative generation constraints in both directions, so unrelated founders may enter later; each connected pedigree starts at the base year. Incompatible matings still raise.

Draw orders. Draw orders are part of the contract, and every change to a seeded output is recorded in the changelog. Results should carry the package version and git revision. The measured evidence in this report is phensim 1.0.0.dev1 at revision `a7371c5`, with NumPy 2.4.6, Numba 0.66.0, msprime 1.4.2 and Python 3.14.6.

Sibling caches. Sibling repositories bind their caches to phensim's source:

- bipred tags cached coalescent segments with a hash of the coalescent, genotype and JIT modules, plus a hand-bumped msprime tag;
- gwfm fingerprints the same files in its simulation cache;
- ldpred3 records a hash of every phensim module in its run archives.

New features that do not touch genotypes therefore live outside those modules. As a further consequence, a change to the coalescent or to the msprime call invalidates downstream caches. Only some of these caches are keyed automatically: in ldpred3, a cache version must be bumped by hand.

12 Limitations and roadmap

Limitations.

- The built-in coalescent models one constant-size population. The msprime split API adds a simultaneous split and an optional admixture pulse; use msprime directly for other demography, migration or selection.
- Block models have no LD across block boundaries.
- The prevalence target is exact only for Gaussian liabilities.
- The RSS oracle assumes a single homogeneous sample.

Roadmap. A survey of the family identified further shared simulators worth consolidating:

- register follow-up records with age of onset, competing mortality and censoring (ltpred, aadgen);
- the exact LD of the AR(1) threshold model (aadgen);

Reproducing this report

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python report/make_evidence.py
cd report && tectonic phensim_report.tex
```

The evidence script is fully seeded. It regenerates every table, figure and quoted measurement, and took 116 s on the recording machine for the saved snapshot. Running it now replaces that snapshot with measurements of the current checkout; compiling the TeX alone preserves the saved evidence.

References

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